AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | AMC

Safe Water
Low Fluoride
Healthy Life

FLUORIDE REMOVAL PLANT

Reliable Solution for Safe & Healthy Drinking Water



- Reduces Fluoride (F-)
- Safe for Drinking
- Natural Adsorbents (Activated Alumina)
- Automatic Operation
- Low Maintenance
- Cost Effective

TYPES OF FLUORIDE REMOVAL PLANTS

FRP FLUORIDE REMOVAL PLANT



- Corrosion resistant FRP vessels
- Compact and lightweight
- Suitable for small to medium capacities
- Easy installation
- Cost effective solution

MSEP FLUORIDE REMOVAL PLANT (Mild Steel Epoxy Painted)



- Mild Steel Epoxy Painted vessels
- Suitable for medium to large capacities
- High strength and long life
- Epoxy internal coating for corrosion protection
- Custom design as per site requirement

MSRL FLUORIDE REMOVAL PLANT (Mild Steel Rubber Lined)



- Mild Steel Rubber Lined vessels
- Suitable for high fluoride & aggressive water
- Rubber lining for superior corrosion resistance
- Ideal for industrial applications
- Long service life

PROCESS FLOW DIAGRAM (WITH BACKWASH SYSTEM)



APPLICATIONS



Residential Societies & Apartments | Rural & Community Water Supply | Schools & Educational Institutions | Hospitals | Hotels & Resorts | Industries | Villages & Government Water Supply Schemes

TECHNICAL SPECIFICATIONS & STANDARD MODELS

Model	Capacity (m³/hr)	Vessel Type (FRP / MSEP / MSRL)	Vessel Size (Dia × Ht in mm) (Per Vessel)	Media Quantity (Activated Alumina) (Kg per vessel)	Service Flow (m³/hr)	Backwash Flow (m³/hr)	Inlet/Outlet Pipe Size (NB)
FRP/MSEP/MSRL-0.5	0.5	FRP / MSEP / MSRL	350 × 1650	50	0.5	1.0	25
FRP/MSEP/MSRL-1	1	FRP / MSEP / MSRL	400 × 1800	75	1.0	2.0	25
FRP/MSEP/MSRL-2	2	FRP / MSEP / MSRL	500 × 2000	125	2.0	4.0	40
FRP/MSEP/MSRL-3	3	FRP / MSEP / MSRL	600 × 2000	175	3.0	6.0	50
FRP/MSEP/MSRL-5	5	FRP / MSEP / MSRL	750 × 2000	300	5.0	10.0	65
FRP/MSEP/MSRL-10	10	MSEP / MSRL	1000 × 2200	550	10.0	20.0	80
FRP/MSEP/MSRL-15	15	MSEP / MSRL	1200 × 2200	800	15.0	30.0	100
FRP/MSEP/MSRL-20	20	MSEP / MSRL	1400 × 2400	1100	20.0	40.0	100
FRP/MSEP/MSRL-30	30	MSEP / MSRL	1600 × 2400	1500	30.0	60.0	150
FRP/MSEP/MSRL-50	50	MSEP / MSRL	2000 × 2500	2300	50.0	100.0	200

Note: 1. Above models are standard. Custom capacities and configurations available on request. 2. Media: Activated Alumina (High Grade).

INLET & OUTLET WATER QUALITY (TYPICAL)

Parameter	Typical Inlet (Ground Water)	Typical Outlet (after Fluoride Removal)
Fluoride (F-)	1.0 – 10.0 mg/L	< 1.0 mg/L
pH	6.5 – 8.5	6.5 – 8.5
TDS	Up to 2000 mg/L	No change
Total Hardness	Up to 600 mg/L	No change
Turbidity	Up to 5 NTU	< 1 NTU
Iron (Fe)	0.1 – 2.0 mg/L	< 0.3 mg/L
Manganese (Mn)	0.1 – 1.0 mg/L	< 0.1 mg/L

KEY FEATURES

- Effective removal of fluoride up to 10 mg/L
- Uses high grade activated alumina media
- Available in FRP, MSEP & MSRL vessels
- Automatic / manual operation options
- Backwash system with pump, blower (optional) and piping
- PLC based control panel with HMI (optional)
- Compact design with skid mounting
- Suitable for borewell, ground water and industrial application
- Low operating cost and easy maintenance
- Custom design as per water analysis and client requirement

OPTIONAL EQUIPMENT

- Raw water pump set
- Pressure sand filter (for turbidity removal)
- Activated carbon filter (for organic removal)
- Air blower / air compressor (for aeration)
- Automatic control valve (multiport/pneumatic)
- Chemical dosing system (if required)
- Online fluoride analyzer (optional)
- Treated water storage tank

PROJECT PHOTOGRAPHS



MSRL Fluoride Removal Plant – 20 m³/hr



FRP Fluoride Removal Plant – 10 m³/hr



MSEP Fluoride Removal Plant – 30 m³/hr



FRP Fluoride Removal Plant – 15 m³/hr



MSRL Fluoride Removal Plant – 40 m³/hr



MSEP Fluoride Removal Plant – 25 m³/hr



FRP Fluoride Removal Plant – 25 m³/hr



FRP Fluoride Removal Plant – 10 m³/hr

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Office & Factory :
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FLUORIDE REMOVAL PLANT

Site Execution Across India

Safe Water | Healthy People | Sustainable Communities

Fazilka, Punjab



SITE IN PROGRESS

FRP Fluoride Removal Plant – 10 m³/hr
Raw Water Fluoride : 3.5 mg/L
Treated Water Fluoride : < 1.0 mg/L

Bathinda, Punjab



SITE COMPLETED

MSEP Fluoride Removal Plant – 20 m³/hr
Raw Water Fluoride : 4.2 mg/L
Treated Water Fluoride : < 1.0 mg/L

Gaya, Bihar



SITE COMPLETED

FRP Fluoride Removal Plant – 15 m³/hr
Raw Water Fluoride : 2.8 mg/L
Treated Water Fluoride : < 1.0 mg/L

Muzaffarpur, Bihar



SITE IN PROGRESS

MSRL Fluoride Removal Plant – 30 m³/hr
Raw Water Fluoride : 5.0 mg/L
Treated Water Fluoride : < 1.0 mg/L

Jodhpur, Rajasthan



SITE COMPLETED

FRP Fluoride Removal Plant – 25 m³/hr
Raw Water Fluoride : 4.5 mg/L
Treated Water Fluoride : < 1.0 mg/L

Bikaner, Rajasthan



SITE IN PROGRESS

MSEP Fluoride Removal Plant – 40 m³/hr
Raw Water Fluoride : 3.8 mg/L
Treated Water Fluoride : < 1.0 mg/L

Ajmer, Rajasthan



SITE COMPLETED

MSRL Fluoride Removal Plant – 50 m³/hr
Raw Water Fluoride : 6.0 mg/L
Treated Water Fluoride : < 1.0 mg/L

Patna, Bihar



SITE COMPLETED

FRP Fluoride Removal Plant – 10 m³/hr
Raw Water Fluoride : 2.5 mg/L
Treated Water Fluoride : < 1.0 mg/L

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Safe Water
Low Fluoride
Healthy People
Better Life

FLUORIDE REMOVAL PLANT

Advanced Adsorption Media for Safe & Healthy Drinking Water



- Removes Fluoride (F-)
- Safe & Healthy Drinking Water
- Natural Adsorbents (Activated Alumina)
- Automatic/ Manual Operation
- Low Maintenance
- Cost Effective

FLUORIDE REMOVAL MEDIA

ACTIVATED ALUMINA



- High adsorption capacity for fluoride
- Standard grade for drinking water treatment
- Granular, spherical form
- Regenerable (with NaOH)
- Suitable for FRP, MSEP & MSRL vessels

FLUORIDE SELECTIVE MEDIA



- Specially formulated media for fluoride adsorption
- Effective at near neutral pH
- High removal efficiency
- Suitable for groundwater and community systems
- Available in regenerable and disposable grades

BONE CHAR MEDIA



- Calcium-phosphate based adsorbent
- Effective fluoride removal
- Used in specific applications
- Replacement as per usage
- Certified drinking-water grade material

SUPPORTING FILTER MEDIA



- Graded Silica Sand
- Pebbles & Gravels
- Anthracite (optional)
- Activated Carbon (for colour, taste & odour)
- Support bed for uniform flow and better performance

TYPICAL PROCESS FLOW DIAGRAM



PLANT CONSTRUCTIONS AVAILABLE



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- Corrosion resistant FRP vessels
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Clean Water for a Fluoride-Free Future

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FRP/MSEP/MSRL-50	50	2000 × 2SRL	2000 × 2500	2300	50.0	100.0	200

Note: 1. Above models are standard. Custom capacities and configurations available on request. 2. Media quantity is indicative and may vary based on raw water quality, pH and design.

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Parameter	Typical Inlet (Ground Water)	Typical Outlet (After Treatment)
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SITE EXECUTION PHOTOGRAPHS



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MSEP Fluoride Removal Plant – 20 m³/hr



MSRL Fluoride Removal Plant – 30 m³/hr



FRP Fluoride Removal Plant – 15 m³/hr



FRP Fluoride Removal Plant – 25 m³/hr



MSEP Fluoride Removal Plant – 40 m³/hr



MSRL Fluoride Removal Plant – 50 m³/hr



FRP Fluoride Removal Plant – 10 m³/hr

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